

19L720-PROJECT WORK 1

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Colorimetric Analysis of Sweat for Sodium monitoring with Integrated User Interface

Abstract

This work presents a novel strip-based colorimetric sensing platform for the detection of sodium in human sweat. The system integrates a photodiode-assisted optical module with precisely aligned LED–photodiode geometry to enable automated, quantitative color analysis, eliminating subjective interpretation. A compact hardware–software interface supports real-time data acquisition and user-friendly visualization, ensuring accurate and consistent biochemical monitoring. Designed for low-cost scalability, the platform offers a non-invasive, portable solution for early detection of electrolyte imbalance and hyperuricemia risk. By bridging laboratory-grade precision with wearable or point-of-care usability, this approach advances personalized and preventive healthcare through continuous sweat-based diagnostics.

Objective

Integration Challenge

- Combine sodium paper-strip photodiode sensing with colorimetric detection on a unified platform.
- Develop a compact LED-photodiode optical system minimizing cross-interference between assays.
- Maintain quantitative accuracy for each biomarker under real sweat and environmental conditions.

Performance Goals

- Detect sodium within physiological ranges relevant for health monitoring.
- Achieve high sensitivity and specificity, comparable to clinical laboratory standards.
- Enable rapid, real-time measurement and analysis suitable for continuous monitoring.

Clinical Translation

- Deliver a portable, user-friendly diagnostic device designed for point-of-care use.
- Facilitate non-invasive testing outside specialized labs for timely health assessment and management.

Economic Impact

- Replace costly, multi-instrument setups with a single low-cost, integrated platform.
- Leverage simple strip-based assays and photodiode detection for easy scalability and manufacturing.
- Reduce overall diagnostic costs, expanding access to affordable health monitoring in wider communities.

Significance of Sweat

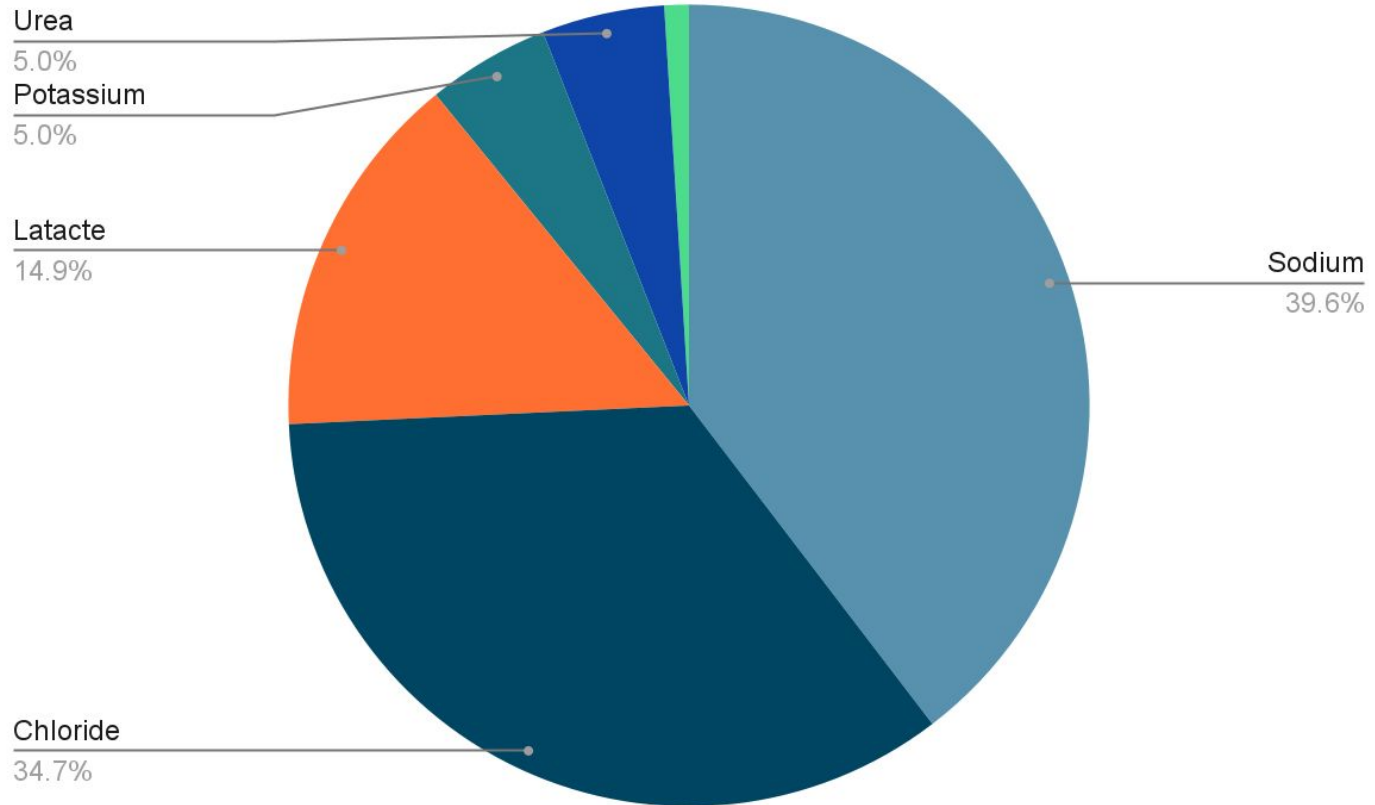
- Non-invasive and Painless: Sweat collection is simple and does not require needles or blood draws, enhancing comfort and compliance.
- Easy Sampling: Sweat can be collected continuously or on-demand using patches or strips, enabling real-time and wearable monitoring.
- Lower Risk: Eliminates risk of infection and complications associated with blood sampling.

Sweat composition

- Water (H₂O): ~9,950 mg (or 9.95 mL)
- Total Dissolved Solutes: ~50 mg

Component	Typical Mass in a 10 mL Sample	Analogy / To Put it in Perspective...
Chloride (Cl ⁻)	~9.2 mg	
Lactate	~13.5 mg	Can be highly variable. May be less at rest.
Sodium (Na ⁺)	~12.4 mg	This is our target. A single grain of table salt weighs about 0.3 mg, so this is like ~30 grains of salt.
Urea	~3.0 mg	A significant waste product.
Potassium (K ⁺)	~2.0 mg	About one-fifth the amount of sodium.
Calcium (Ca ²⁺)	< 0.2 mg	A tiny trace amount.
Magnesium (Mg ²⁺)	< 0.1 mg	A tiny trace amount.
Other Organics	~ a few mg	Amino acids, proteins, etc., make up the rest.
TOTAL	** ~ 40 - 60 mg **	

Sweat composition



Significance of sodium

- **Prevention of sodium imbalance:**

Monitoring sweat sodium helps detect early deviations from normal levels, preventing dangerous conditions like hypernatremia or hyponatremia.

- **Early health warning:**

Since sweat composition reflects blood electrolyte balance, changes in sweat sodium can serve as an **early warning signal** for dehydration or electrolyte disorders.

- **Neurological and muscular safety:**

Regular tracking helps avoid sodium fluctuations that cause **nerve dysfunction, muscle cramps, and brain swelling/shrinkage**.

- **Continuous, non-invasive monitoring:**

Sweat sensing offers a **real-time and painless** way to monitor sodium balance, unlike traditional blood sampling.

Significance of sodium

(*contd...*)

High Prevalence of Sodium Imbalances:

- **Hyponatremia (Low Sodium):** This is the most common electrolyte disorder observed in clinical settings, affecting up to 30% of hospitalized patients. It is associated with a range of conditions, including heart failure, kidney disease, and diuretic use. The mortality rate for patients with severe hyponatremia can be as high as 50%.
- **Hypernatremia (High Sodium):** While less common, hypernatremia is a serious condition, often resulting from dehydration, particularly in the elderly and critically ill. It carries a high mortality rate, ranging from 40% to 60%, emphasizing the need for early detection and intervention.
- **Impact on High-Risk Populations:**
 - **Elderly:** This demographic is particularly vulnerable to both hyponatremia and hypernatremia due to age-related physiological changes, multiple comorbidities, and polypharmacy.
 - **Individuals with Chronic Illnesses:** Patients with conditions like congestive heart failure, chronic kidney disease, and cystic fibrosis require careful sodium management to prevent complications and hospital readmissions.

The development of a non-invasive sodium monitoring device directly addresses this clinical need, offering a way to track at-risk individuals outside of the hospital, enabling preventative care and reducing healthcare costs.

Study / Source	Sample Size / Population	Condition Studied	Key Mortality Findings
Whelan et al., (PLOS ONE)	27,889 hospitalized adults	Hyponatremia & Hypernatremia	Normal sodium: 5% in-hospital deaths; mild hyponatremia: 9% ; severe hyponatremia: 14% ; mild hypernatremia: 28% ; severe hypernatremia: 52% mortality.
Corona et al., (Meta-analysis, 81 studies)	Multicenter (various conditions)	Hyponatremia	Patients with hyponatremia had a 2.6× higher risk of death vs. normal sodium levels.
Kovesdy et al., (Nephrol. Dial. Transplant.)	45,333 CKD patients (stages 3–4)	Dysnatremia (both hypo & hyper)	Both low and high sodium linked to significantly increased all-cause mortality after adjustment.
Hoorn et al., (Clin. J. Am. Soc. Nephrol.)	60,944 hospitalized adults	Sodium variation ≥6 mEq/L	Even small fluctuations in sodium correlated with higher hospital and 1-year mortality .
Verbalis et al., (Am. J. Med.)	Hospitalized patients with severe hyponatremia (<120 mEq/L)	Severe Hyponatremia	High fatality due to neurological complications (cerebral edema, seizures) and multi-organ failure.

Study / Source	Population	Key Findings
KIMS, Kerala	5,558 in-patients	Hypernatremia (175 pts) → 12% mortality
MICU, Udaipur	798 ICU pts	Hypernatremia → 84% deaths; Hyponatremia → 54% deaths (high-risk patients)
George Institute / ICMR	National dietary survey	High salt intake → ~175,000 deaths/year via CVD & CKD
Salt intake modelling	Urban Indians	Reducing sodium to WHO levels could prevent ~300,000 deaths/10 yrs

Proposed Methodology

- Collect sweat samples using absorbent pads or direct test strip contact.
- Detect sodium via colorimetric strips, both producing color changes based on concentration.
- Employ a dedicated LED–photodiode optical module to illuminate strips under controlled conditions and measure reflected light intensity.
- Digitize and process signals using calibration models developed from synthetic sweat standards for accurate biomarker quantification.
- Classify processed values against physiological ranges to assess normal or abnormal levels.
- Integrate results into a user-friendly digital interface providing real-time feedback and actionable insights.
- Validate system performance against laboratory reference methods to ensure accuracy and clinical relevance.

Significance of Colorimetry

- **Highly Sensitive & Accurate:** Detects even small fluctuations in sodium concentration critical for health monitoring.
- **Non-Invasive & Real-Time:** Enables continuous tracking from sweat without needles or lab testing.
- **Simple Quantification:** Color intensity directly correlates with sodium levels — easy to interpret and analyze.
- **Low-Cost & Portable:** Requires minimal instrumentation; ideal for wearable or field applications.
- **Smart Integration:** Compatible with smartphone or IoT-based optical sensing systems for digital readouts.
- **Reliable Chemistry:** Uses stable reagents and proven principles for consistent, repeatable results.
- **Bridges Lab & Real-Life Use:** Transforms clinical-grade precision into practical, on-body health monitoring.

Novelty

- **Dynamic Environmental Integration:**

Combines sodium colorimetry with live temperature/humidity sensing, accounting for environmental effects on sweat sodium—an aspect widely proven to impact readings, but nearly absent in published wearable systems.

- **Contextualized Reporting & Advice:**

Moves beyond raw reporting by interpreting sodium loss relative to ambient conditions, providing situationally-aware feedback (hydration, warning alerts) not currently available in most commercial or research biosensors.

- **Advanced Analytics for Personalization:**

Employs data modeling and, if possible, machine learning to adapt recommendations based on individual, activity, and environmental factors—closing a recognized research gap for hydration and risk prediction.

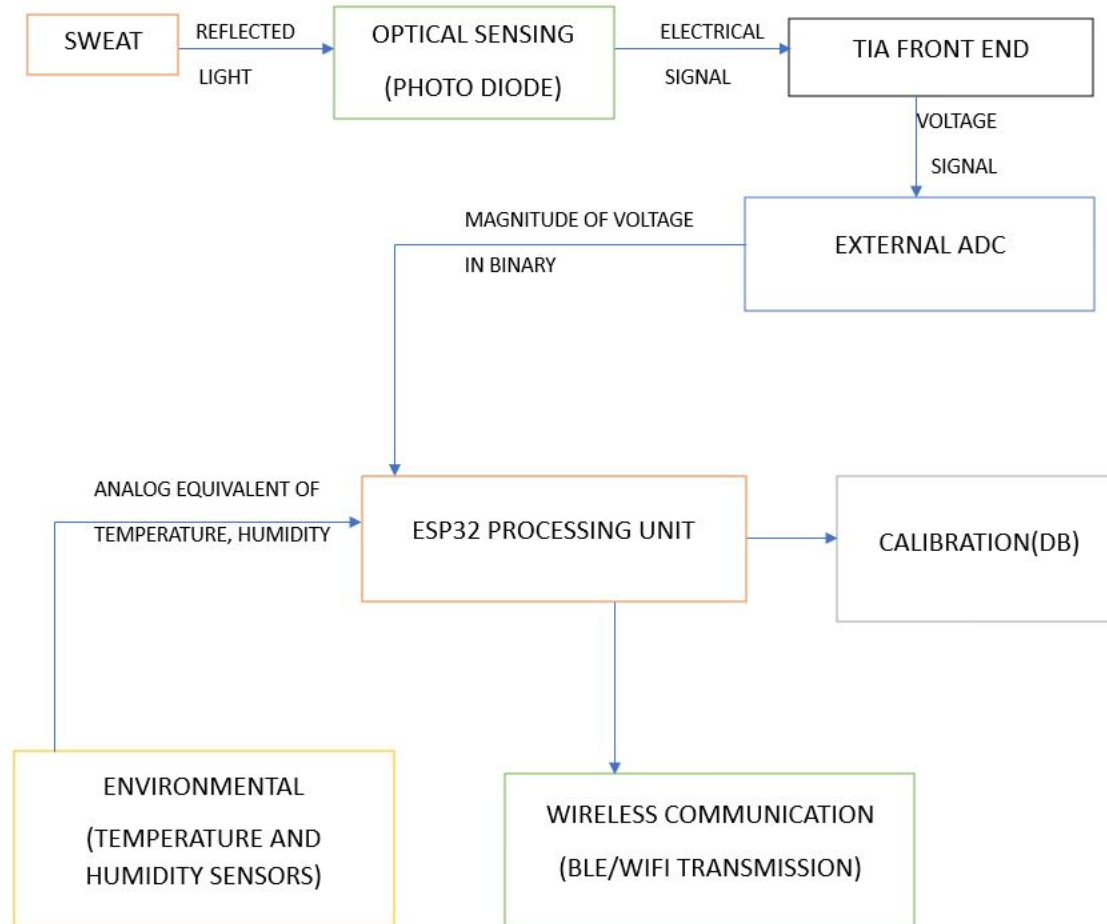
- **Recent studies:**

- Sweat sodium varies substantially with environment, activity, and personal traits—ignoring this risks misreporting.
- Most current devices and reviews call for contextual analytics and better real-world interpretation.
- No mass-market or published wearable adjusts sodium recommendations in real-time with ambient factors.

Feasibility Analysis

- **Feasibility:** Uses low-cost, readily available components (ESP32, photodiode, LEDs, AFEs like ADS1220/LMP91000), ensuring reliable multimodal sensing within budget and time.
- **Processing & Communication:** ESP32 provides sufficient computation, real-time data acquisition, preprocessing, and wireless communication (Wi-Fi/BLE).
- **Scope (Immediate):** Design and implement a multimodal biosensing platform integrating optical and electrochemical sensors, validate with model biomarkers, and visualize results on a wireless dashboard.
- **Scope (Extended):** Future expansion with additional sensing modalities, machine learning for decision support, miniaturization into point-of-care devices, and cloud-based remote monitoring.

MULTI MODAL BIOSENSING PLATFORM ARCHITECTURE



Need for Temperature & Humidity Sensing

Temperature & Humidity:- To Control for Sweat Sample Evaporation (The Biggest Error Source)
To Ensure Consistent Chemical Reaction Speed

Reason:

1. **Error Flagging:** The device can log the temperature and humidity with every reading. If a user gets an unexpectedly high sodium value, you can check the environmental data. If it was a hot, dry day, you have a probable cause for the anomaly.
2. **Advanced Correction:** A highly advanced version of your device could use the temp/humidity data to apply a **correction factor**. Using a pre-determined model, it could slightly adjust the final reading downwards to compensate for the estimated evaporative loss.

Working

Kubelka-Munk (K-M) theory is a model that describes how light behaves when it hits a surface that is **both light-absorbing and light-scattering**. This is exactly what your test strip is: an opaque surface where a colored substance absorbs light, and the base material (e.g., paper or a chemical matrix) scatters it in all directions.

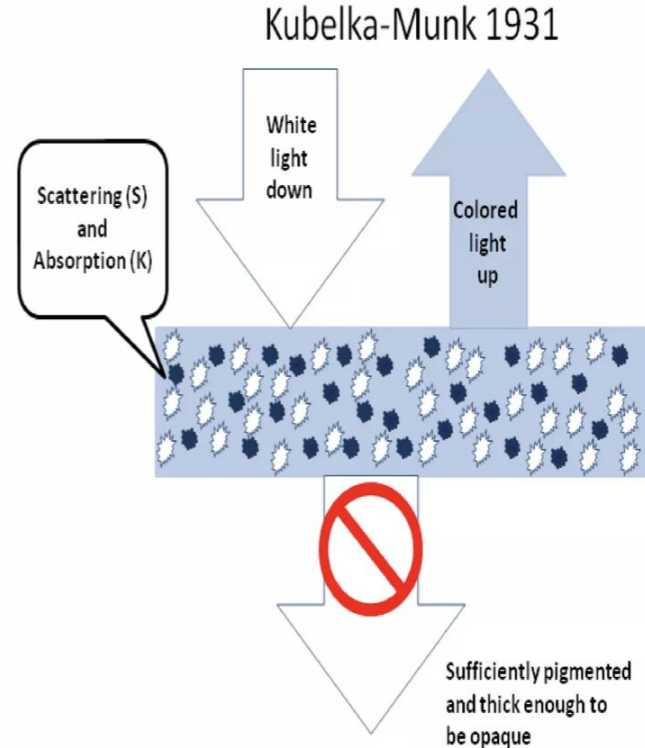
The Kubelka-Munk Equation

After a lot of calculus, the theory produces a wonderfully practical equation that connects the measurable reflectance of the surface to the K and S coefficients. The most common form is:

$$K / S = (1 - R)^2 / 2R$$

Where:

- **R** is the **reflectance** of your sample, which is the value your photodiode measures. (Specifically, it's the ratio of reflected light intensity to incident light intensity, a value between 0 and 1).
- **K** is the absorption coefficient.
- **S** is the scattering coefficient.

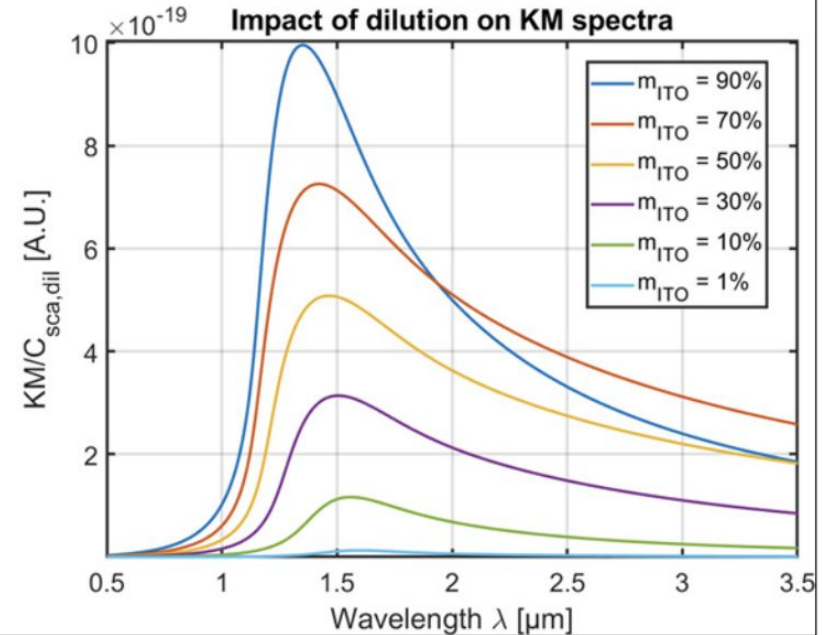


Working

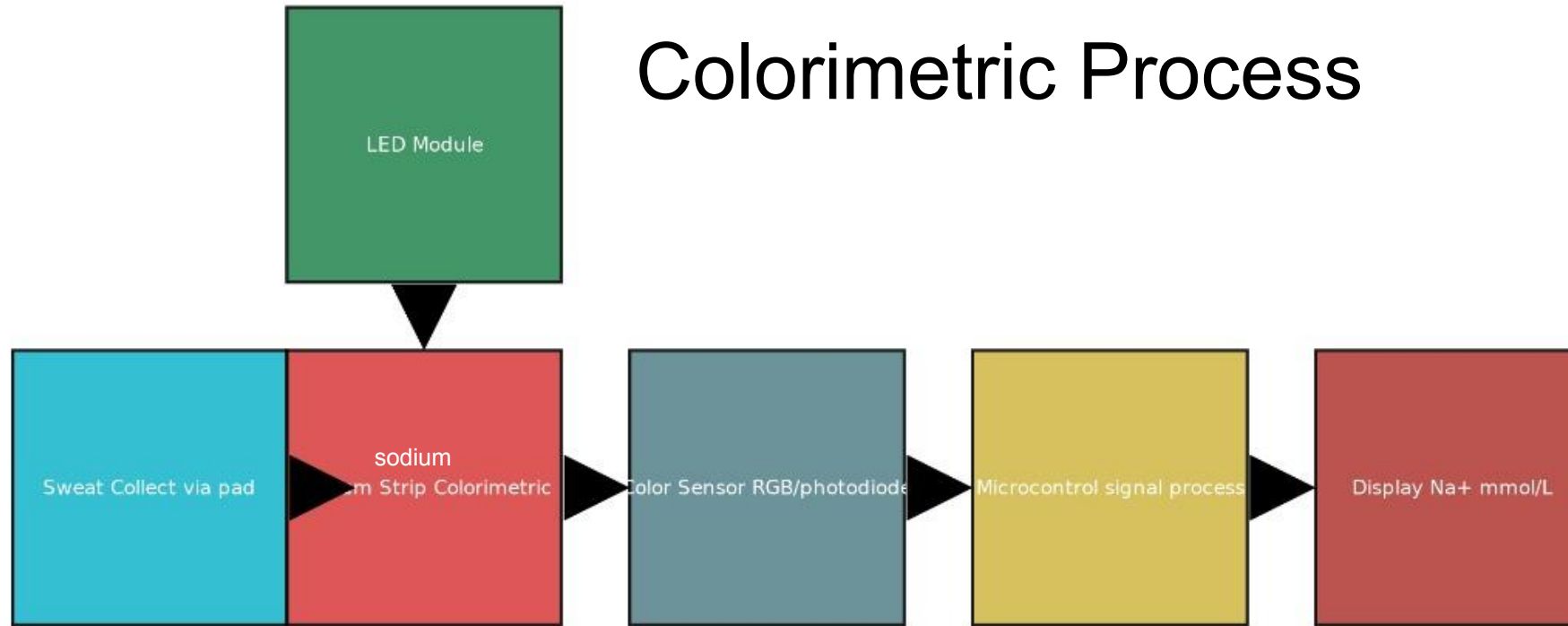
Direct Proportionality: The most important takeaway is that the **height of the Kubelka-Munk (KM) peak is directly proportional to the concentration of the substance**. The 90% sample has the highest peak, the 70% sample has the next highest, and so on, all the way down to the 1% sample. This visually proves that the K-M value is a reliable indicator of concentration.

- **X-Axis (Wavelength):** This shows the different wavelengths (colors) of light being shone on the sample, from visible to near-infrared (0.5 to 3.5 μm).
- **Y-Axis (KM/C_{sca,dil}):** This is the calculated **Kubelka-Munk value**, which is derived from the reflectance measurement (R^∞). Think of this as the K-M equivalent of "Absorbance"—it is directly proportional to the concentration of the substance you're interested in.
- **The Curves ($m_{\text{ITO}} = \%$):** Each colored line represents a different sample with a known concentration (mass fraction, m) of a substance called Indium Tin Oxide (ITO). The blue line is 90% ITO (very concentrated), and the light blue line is 1% ITO (very diluted).

graph with "Sodium Concentration" on the X-axis and the "Height of the Blue Peak (K/S value)" on the Y-axis, we would get a straight line—calibration curve.



Colorimetric Process



Progress

I. Core System & Connectivity

- Developed a robust firmware for the **ESP32 microcontroller** to act as the central processing unit.
- Implemented **automatic WiFi connectivity** with a non-blocking auto-reconnect manager to ensure stable network presence.
- Hosted a lightweight **web server directly on the ESP32** to serve live data on demand.
- Established a **REST API** for seamless, real-time communication between the ESP32 and the web interface (/api/sensors, /api/status, /api/test).

II. Web-Based User Interface (The "Wireless Dashboard")

- Designed and built a dynamic, **web-based dashboard** using HTML, CSS, and JavaScript for user-friendly data visualization.
- Enabled **live data display** of all sensor values with a configurable update frequency (currently set to every 1 second).
- Integrated a **system log panel** to display real-time status messages, connection events, and hardware errors from the ESP32.
- Implemented a "heartbeat" system in the web interface to **automatically detect connection loss** and update the UI in real-time.

Progress contd

1. Sample & User Errors

These occur before light hits the strip and cause the most inaccuracy.

- Contamination:** Oils, lotions, or dirt interfere with reactions or block light.

- Inconsistent Volume:** Too little sweat → incomplete reaction → low reading.

- Evaporation:** Delay before testing → concentrated sweat → high reading.

- Timing Errors:** Reading too early or late gives wrong results.

- Air Bubbles:** Prevent full contact, causing low readings.

2. Optical Errors

Affect light transmission between LED, strip, and photodiode.

- Ambient Light Leakage:** External light lowers measured concentration.

- Strip Misalignment:** Changes reflected light intensity.

- Dirty Optics:** Dust or smudges cause inconsistent readings.

- Surface Defects:** Wrinkles or gloss alter reflectance.

3. Chemical & Strip Errors

Related to strip quality and handling.

- Poor Storage:** Heat or humidity degrades reagents → weak color.

- Batch Variation:** Slight reagent differences shift calibration.

4. Electronic Errors

Originating from components.

- LED Drift:** Brightness changes with temperature/age; recalibrate often.

- Noise:** Random voltage fluctuations; average multiple readings.

- Photodiode Temperature Effect:** Slight sensitivity drift.

5. Systematic & Calibration Errors

Due to setup and assumptions.

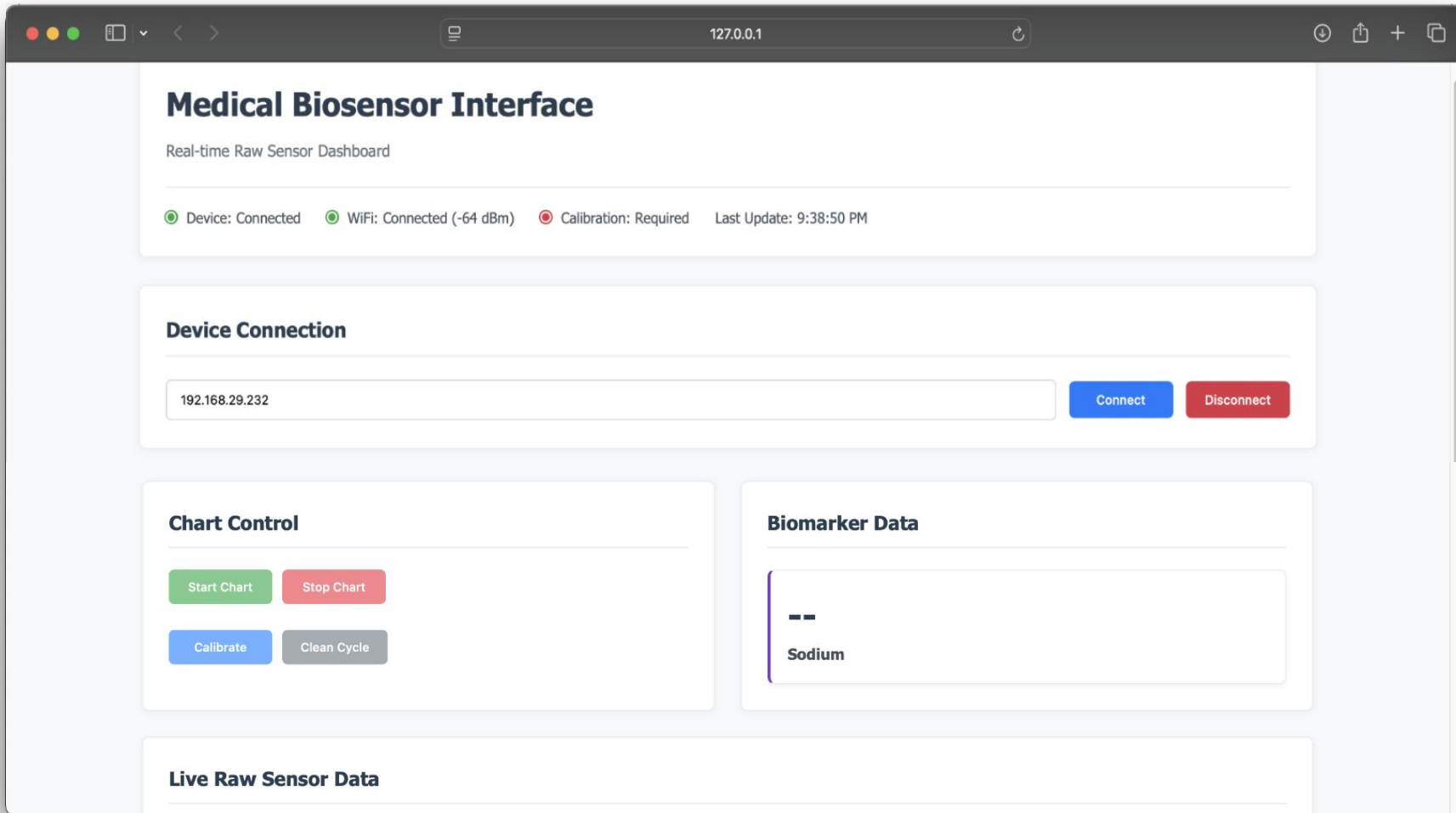
- Poor Calibration:** Dirty or wrong reference skews all results.

- Non-Linearity:** Linear fits fail at extreme concentrations.

Progress contd

Real-World Challenges to Solve

- A. Challenge: Sample Contamination
 - **The Problem:** Sweat isn't pure. It can be contaminated with skin oils, lotions, dirt, or old sweat that has dried on the skin. These contaminants can interfere with the chemical reaction or change the optical properties of the strip, leading to false readings.
- B. Challenge: Consistent Sample Volume
 - **The Problem:** The amount of sweat applied to the strip can affect the final color intensity. A larger drop might result in a darker color, even if the concentration is the same. You cannot use a pipette to apply sweat.
- C. Challenge: Evaporation
 - **The Problem:** Sweat samples are tiny. They can evaporate very quickly, especially in a warm environment. As the water evaporates, the sodium concentration in the remaining liquid skyrockets, which will give you a falsely high reading.
- D. Challenge: Chemical Interference
 - **The Problem:** Sweat contains other electrolytes besides sodium, primarily potassium (K^+), chloride (Cl^-), and calcium (Ca^{2+}).



Medical Biosensor Interface

Real-time Raw Sensor Dashboard

● Device: Connected ● WiFi: Connected (-64 dBm) ● Calibration: Required Last Update: 9:38:50 PM

Device Connection

192.168.29.232

Connect

Disconnect

Chart Control

Start Chart

Stop Chart

Calibrate

Clean Cycle

Biomarker Data

--

Sodium

Live Raw Sensor Data

Live Raw Sensor Data

--

Colorimetric (AU)

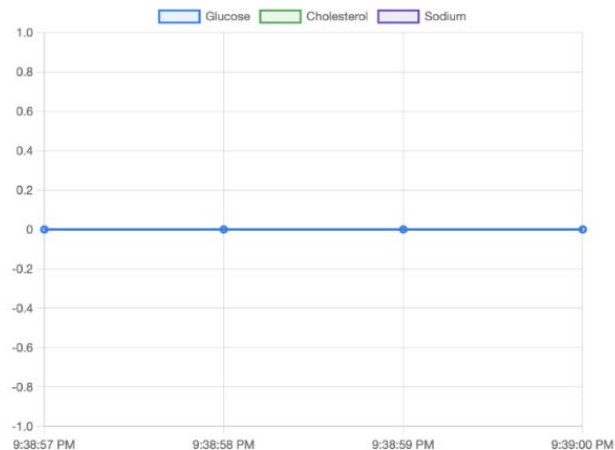
30.3

Temperature (°C)

79.0

Humidity (%)

Real-time Chart

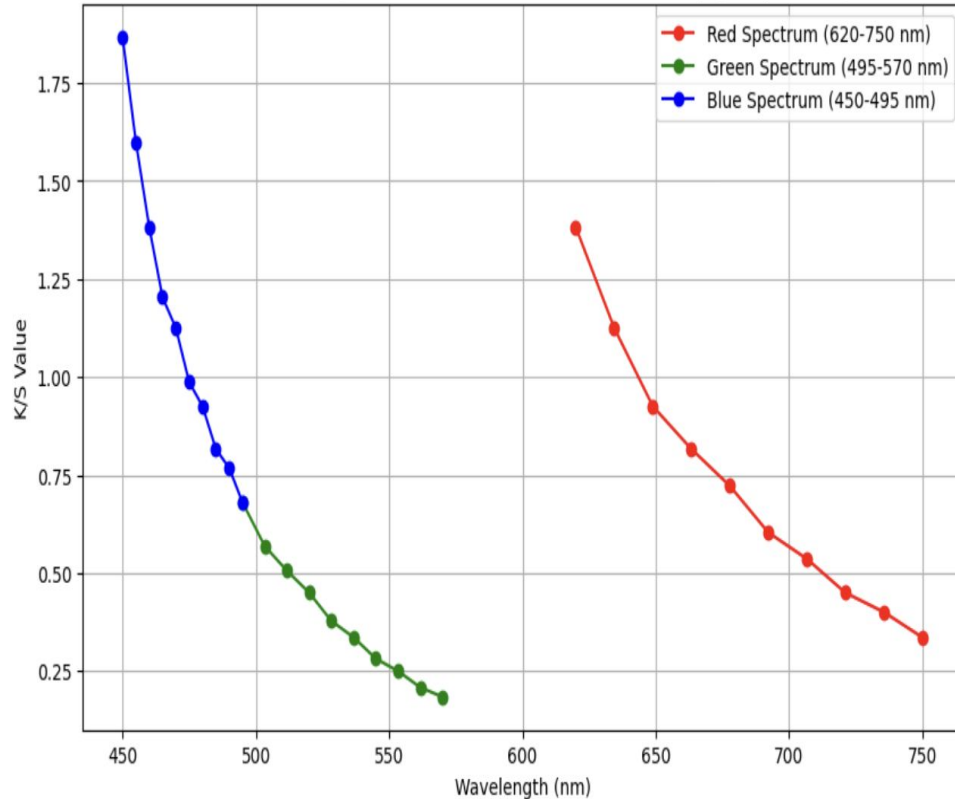


System Log

```
9:14:06 PM: Failed to fetch sensor data: Load failed
9:16:15 PM: Disconnected by user.
9:16:16 PM: Connecting to http://192.168.29.232...
9:17:01 PM: Connecting to http://192.168.29.232...
9:17:13 PM: Connecting to http://192.168.29.232...
9:17:28 PM: Failed to fetch sensor data: Load failed
9:19:22 PM: Failed to fetch sensor data: Load failed
9:21:34 PM: Failed to fetch sensor data: Load failed
9:38:43 PM: Connecting to http://192.168.29.232...
9:38:43 PM: Successfully connected! Signal: -67 dBm
9:38:56 PM: Charting started.
9:39:01 PM: Charting stopped.
```

Kubelka-Munk Simulation Model

Kubelka-Munk K/S Values for Sodium Colorimetric Strip (Sample Data)



--- Sodium Level Calculation from Sample RGB Data

RGB Input: (180, 100, 80)

K/S (R, G, B): (0.061, 0.471, 0.751)

Estimated Sodium Level: 197.0 mM

RGB Input: (150, 130, 90)

K/S (R, G, B): (0.144, 0.236, 0.593)

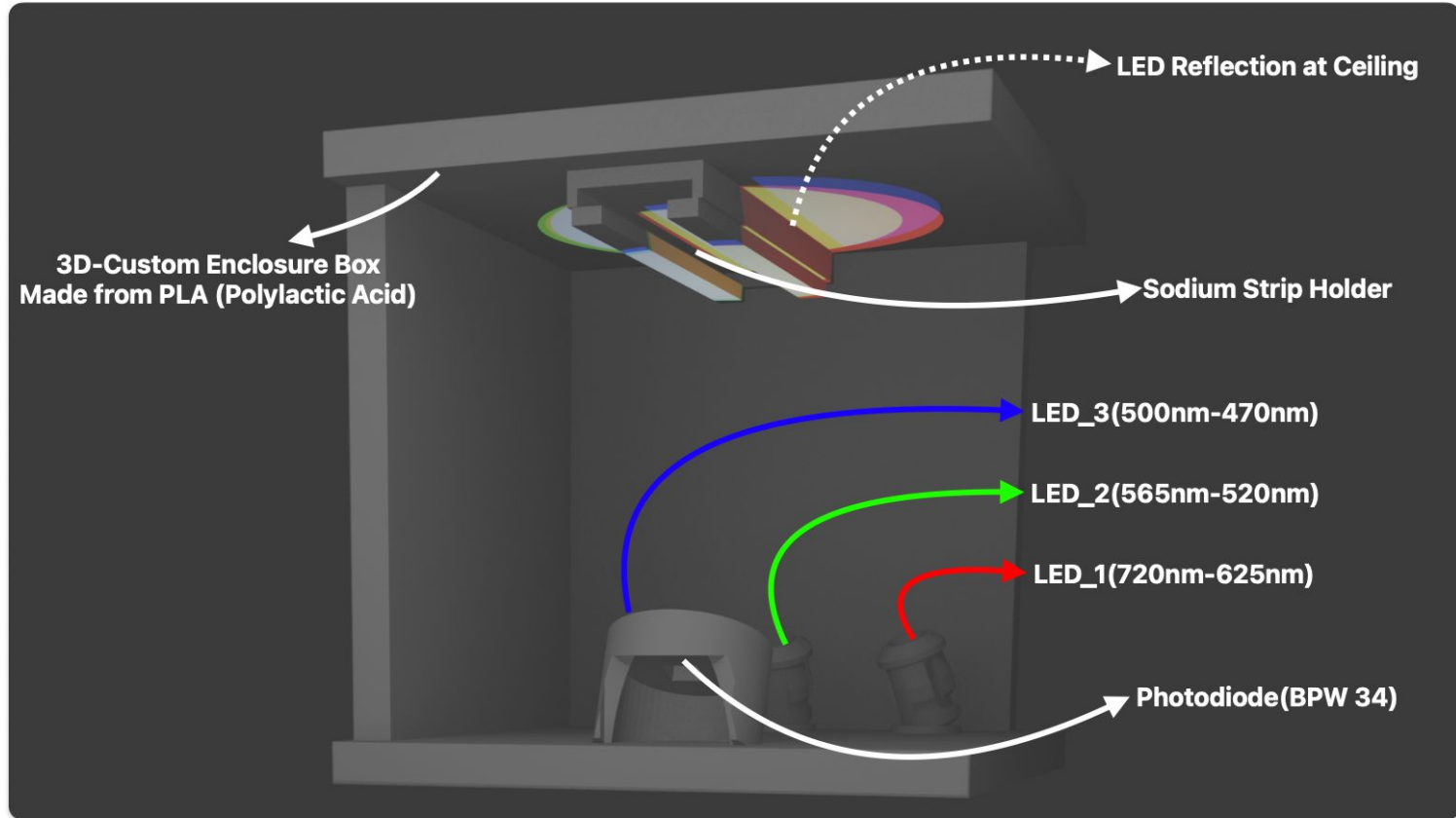
Estimated Sodium Level: 91.1 mM

RGB Input: (100, 130, 170)

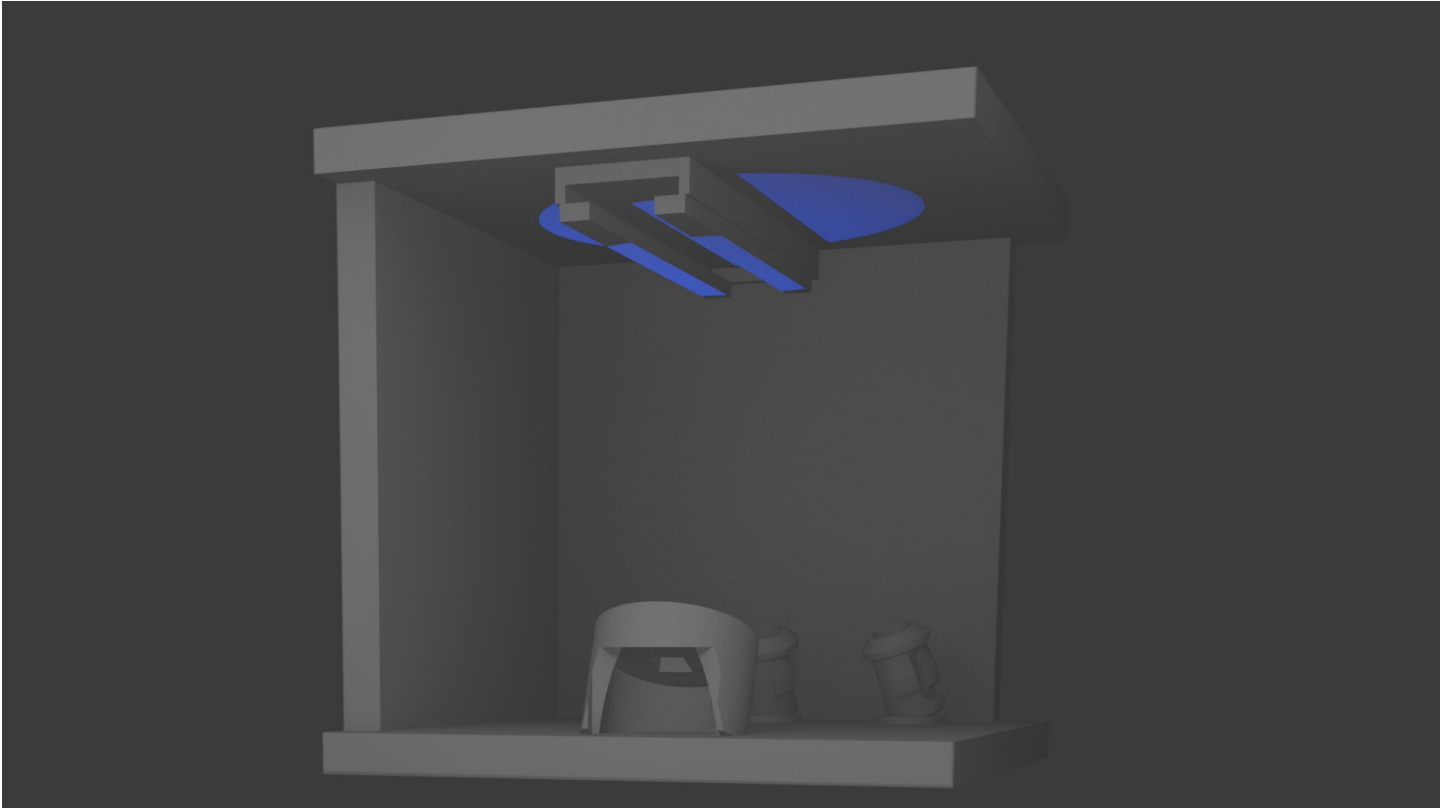
K/S (R, G, B): (0.471, 0.236, 0.083)

Estimated Sodium Level: 91.1 mM

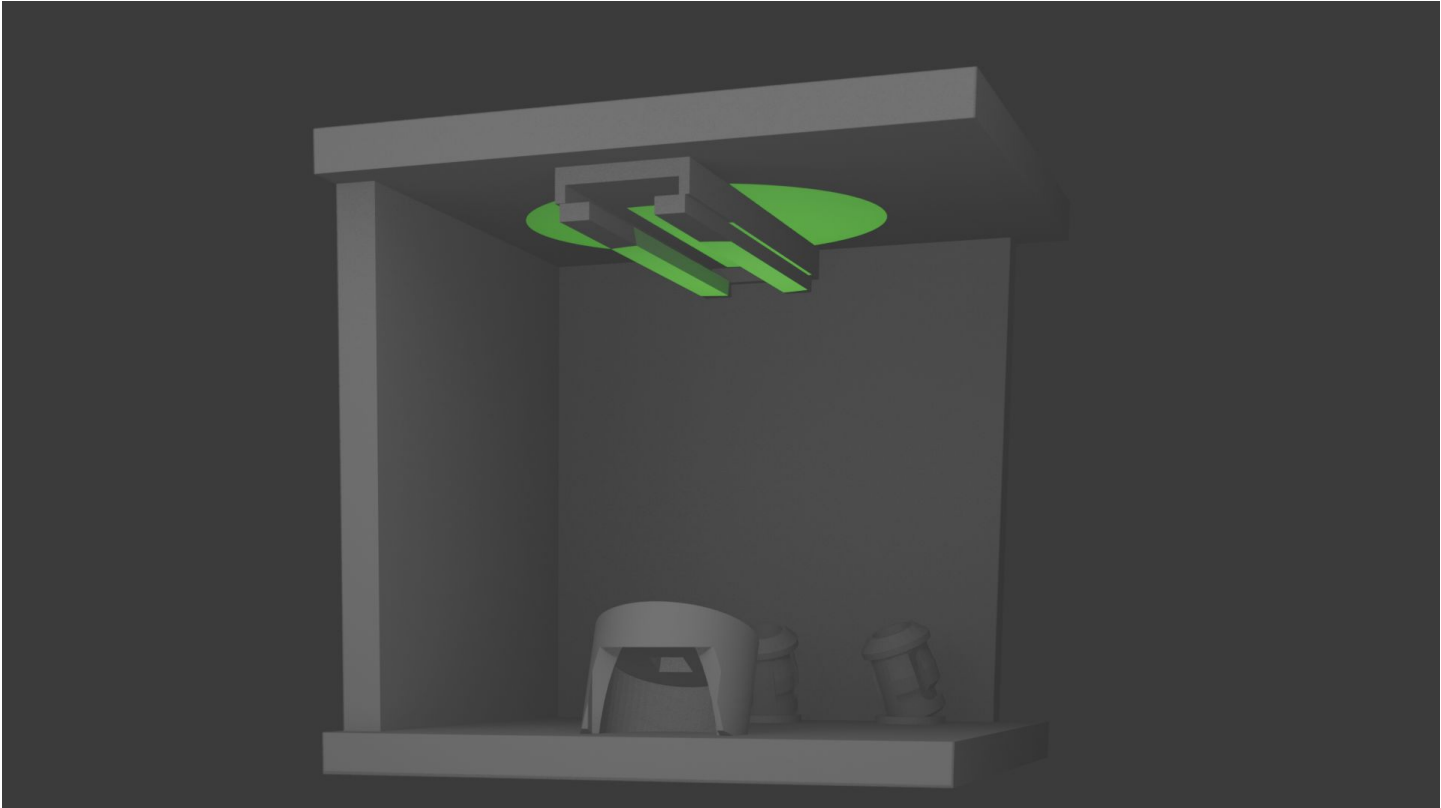
3D-Simulation



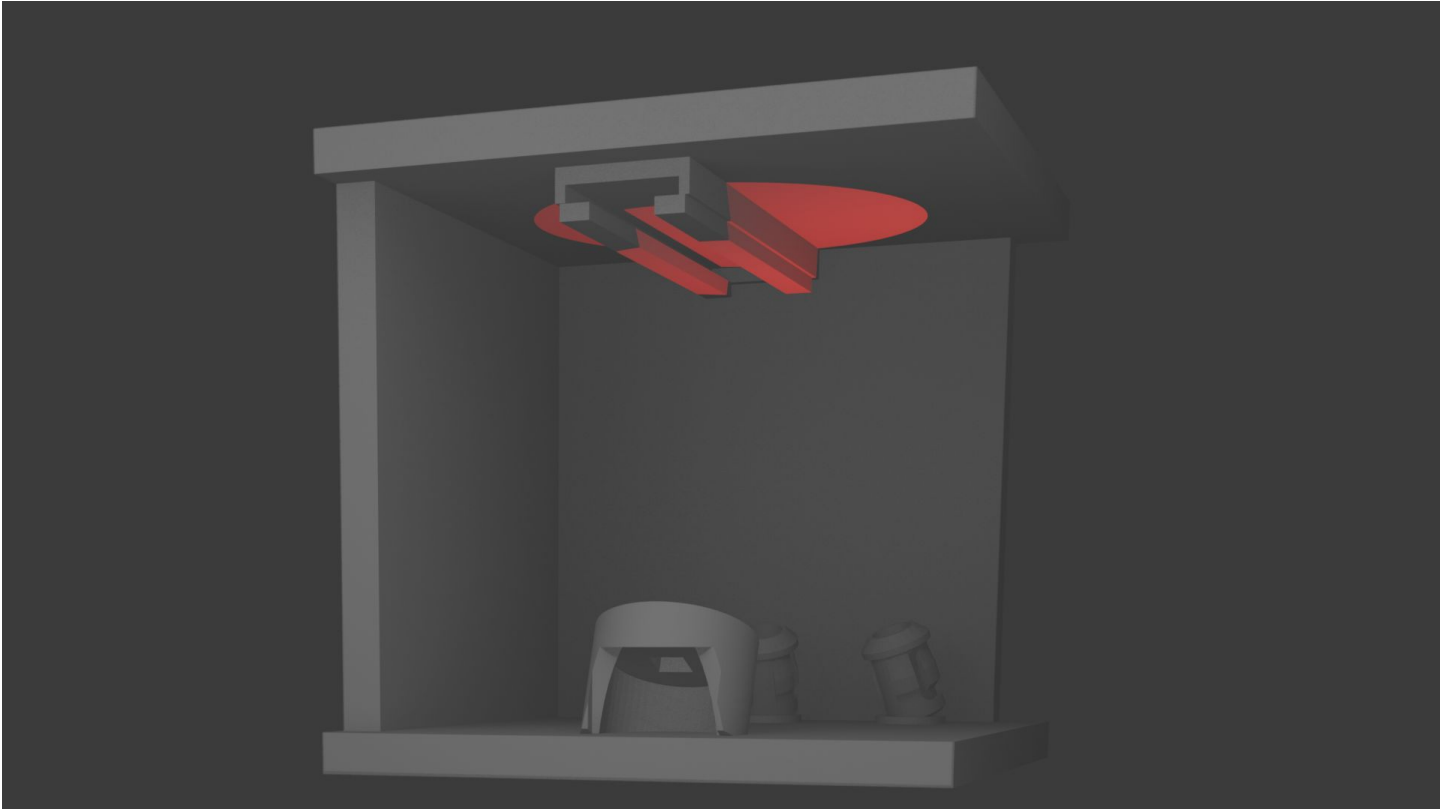
Simulated(Blue LED)



Simulated(Green LED)



Simulated(Red LED)



Literature Survey

1. **Title:** Smartphone Biosensor System with Multi-Testing Unit Based on Localized Surface Plasmon Resonance Integrated with Microfluidics Chip
2. **Title:** Smartphone-Based Electrochemical Systems for Glucose Monitoring in Biofluids: A Review
3. **Title:** Wireless Integrated Biosensors for Point-of-Care Diagnostic Applications

Key Components (Electronics):

- Sensor Platform: Electrochemical biosensors (screen-printed electrodes, paper-based sensors, and CMOS-integrated platforms) for small-molecule, glucose, and pathogen detection.
- Analog Front-End: Compact potentiostat (discrete 3-op-amp system or IC-based like LMP91000) for WE–RE–CE control and current measurement.
- Data Handling: Integrated ADC within the microcontroller (or external high-resolution ADC) for signal acquisition and preprocessing.
Wireless Technology: BLE for smartphone pairing and short-range connectivity; Wi-Fi/USB for extended cloud or PC-based data analysis.
- Microcontroller: ESP32-S3 (dual-core processor with integrated Wi-Fi + BLE for sensing + IoT).

Power Management:

- Rechargeable Li-ion/Li-Po battery with TP4056 charging module and voltage regulator (3.3V/5V).
Optional boost converter for stable supply during wireless communication.

User Interface & Connectivity:

- Smartphone applications (Android/iOS) for real-time monitoring and control.
- PC dashboard for advanced data visualization and processing.
- Optional web/cloud integration for remote diagnostics and long-term storage.

Literature Survey

1. **Title:** Portable Miniaturized IoT-Enabled Point-of-Care Device for Electrochemical Sensing of Zopiclone in Cocktails
2. **Title:** Portable Electrochemical Sensing System Attached to Smartphones and Its Incorporation with Paper-based Electrochemical Glucose Sensor
3. **Title:** Recent developments towards portable point-of-care diagnostic devices for pathogen detection
4. **Title:** Applications of CMOS Devices for the Diagnosis and Control of Infectious Diseases

Key Components (Electronics):

- **Microcontroller:**
 - **Primary Choice: ESP32-S3.** This microcontroller is selected for its powerful dual-core processor, integrated Wi-Fi and Bluetooth LE 5.0, and ample GPIO pins. It can single-handedly manage the electrochemical measurements, process data, and handle wireless communication, reducing complexity and cost.
- **Analog Front-End:**
 - **Potentiostat Circuit:** A custom-designed potentiostat will be implemented using operational amplifiers (e.g., **LMP91000** or a discrete design with **OPA2320**) to accurately control the voltage and measure the resulting current from the electrochemical sensor.
- **Power Management:**
 - A compact **TP4056-based Li-ion/Li-Po battery charging module** will be used for rechargeable power, ensuring portability. A voltage regulator (e.g., **MCP1700**) will provide a stable 3.3V supply to the microcontroller and other components.
- **User Interface & Connectivity:**
 - The primary interface will be a custom smartphone application connected via **Bluetooth Low Energy (BLE)**. The ESP32's built-in BLE capabilities make this straightforward to implement. An optional small OLED display (e.g., **SSD1306**) could be added for device status feedback.

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- Fujimoto, Takuya, et al. "Portable electrochemical sensing system attached to smartphones and its incorporation with paper-based electrochemical glucose sensor." *International Journal of Electrical and Computer Engineering* 7.3 (2017): 1423.
- Forouhi, Saghi, and Ebrahim Ghafar-Zadeh. "Applications of CMOS devices for the diagnosis and control of infectious diseases." *Micromachines* 11.11 (2020): 1003.

Clinical Need for Sodium Reference

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1. Hoorn, E. J., & Zietse, R. (2017). Diagnosis and Treatment of Hyponatremia: Compilation of the Guidelines. *Journal of the American Society of Nephrology*, 28(5), 1340–1349.*
2. Noakes, T. D., Sharwood, K., Speedy, D., et al. (2005). Three independent biological mechanisms cause exercise-associated hyponatremia: evidence from 2,135 weighed competitive athletic performances. *Proceedings of the National Academy of Sciences*, 102(51), 18550-18555.*

On Sodium Imbalances in Athletes (Exercise-Associated Hyponatremia):

1. Hew-Butler, T., Rosner, M. H., Fowkes-Godek, S., et al. (2015). Statement of the Third International Exercise-Associated Hyponatremia Consensus Development Conference, Carlsbad, California, 2015. *Clinical Journal of Sport Medicine*, 25(4), 303–320.*
2. Noakes, T. D., Sharwood, K., Speedy, D., et al. (2005). Three independent biological mechanisms cause exercise-associated hyponatremia: evidence from 2,135 weighed competitive athletic performances. *Proceedings of the National Academy of Sciences*, 102(51), 18550-18555.*

On Sodium Imbalances in High-Risk Clinical Populations (Elderly, Chronic Illness):

1. Mann, D. L., Givertz, M. M., et al. (2022). 2022 AHA/ACC/HFSA Guideline for the Management of Heart Failure. *Journal of the American College of Cardiology*, 79(17), e263–e421.*
2. Rondon-Berrios, H., Argyropoulos, C., Ing, T. S., et al. (2017). Hypertonicity: Clinical entities, manifestations and treatment. *World Journal of Nephrology*, 6(1), 1–13.*
3. Shorofsky, S. R., Luria, R., & Riba, A. L. (2021). The Global Burden of Cystic Fibrosis. *Current Opinion in Pulmonary Medicine*, 27(6), 598-603.*